

LLM-Guided Hierarchical Belief Planning for Online Semantic Exploration and Task Execution

Shuo Zhang¹, Zhongkui Li¹, Meng Guo¹, and Yanran Wei¹

Abstract—Robots executing long-horizon temporal tasks in partially known semantic environments must often search for uncertain targets while making task-execution decisions. Large language models can provide commonsense priors over likely target locations, but target likelihood alone is not enough when observations can change the remaining route and the feasible task order. In such settings, the most likely location may be suboptimal if searching it leads to higher downstream task-completion cost. We propose an LLM-guided hierarchical semantic belief-planning framework that maintains beliefs over uncertain target locations, updates them from observations, and evaluates exploration and execution actions by finite-horizon expected task-completion cost. The planner uses receding-horizon belief-tree search with multi-stage pruning and a terminal cost estimate to support online execution. In hospital-like transport scenarios, the proposed framework reduces execution time and LLM-query overhead compared with representative deterministic-planning and model-based-search baselines, while keeping online planning time practical.

Index Terms—LLM-guided semantic priors, task planning under uncertainty, online belief-space planning, receding-horizon planning.

I. INTRODUCTION

Robots executing long-horizon temporal tasks [1] in partially known semantic environments must often act before all task-relevant facts are observed. In a hospital transport task, for example, a drug or patient chart may be hidden in one of several rooms or receptacles, and the robot must decide whether to search, pick a localized item, or deliver an item already in hand. Object locations in such environments are not arbitrary: they follow semantic regularities that can be estimated from LLM-generated commonsense knowledge [2], [3]. The central difficulty, however, is not only estimating where a target may be. Under temporal task constraints, both exploration and task-execution actions can change the remaining route, the set of admissible task events, and the value of future observations. A useful action therefore cannot be selected only by immediate search likelihood or local travel cost; it should be evaluated by its expected effect on completing the remaining task. Fig. 1 illustrates this coupling in a hospital-like transport scenario.

Existing methods address parts of this problem from two directions. One line of work handles partial observability through closed-loop task or task-and-motion planning, replanning after new information is observed [4]; it can recover from new observations but handles missing information mostly after

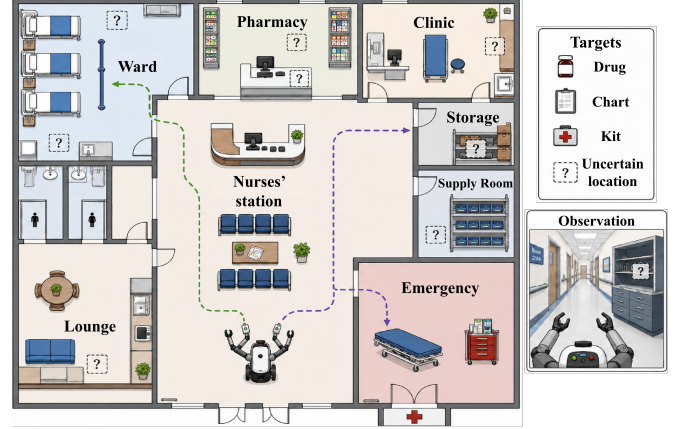


Fig. 1. Problem illustration. A robot executes a temporal transport task in a partially known hospital environment. Target objects may be hidden at uncertain semantic locations.

it is encountered. Another line of work brings environmental uncertainty directly into planning. Information-gathering and semantic object-search methods use observations, object correlations [3], or semantic priors [2] to reduce uncertainty about task-relevant targets. These methods can make search more efficient, but they usually do not model the coupling between search and downstream task execution.

Among methods that bring semantic uncertainty into planning, Yang et al. introduce the EPoG framework, which uses LLM-based situated replanning to infer likely locations for unobserved targets on a scene graph and then performs deterministic planning for exploration and sequential manipulation [5]. However, because EPoG commits to a single likely graph hypothesis before execution, it can favor the most likely target location even when another search choice would yield lower remaining task cost. Paudel et al. quantify target-location uncertainty with LLM-informed priors and Bellman-style search evaluation [2]. Scaling this idea to multi-target temporal tasks requires reasoning over joint target beliefs and task progress; treating targets independently is tractable but remains target-local, missing effects on other unresolved targets, the remaining route, and precedence constraints. Kim et al. further model target-location uncertainty with LLM-generated hierarchical priors and co-location cues [3]. These estimates make search more informed, yet the action value is still tied mainly to where targets are likely to be found, rather than to how a search or execution decision changes the remaining temporal-task cost.

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We therefore propose an LLM-guided hierarchical belief-planning framework for online semantic exploration and task execution. The framework maintains an updated hierarchical belief over uncertain target locations and reasons about how this uncertainty affects the remaining task. At each decision step, it evaluates both search and task-execution actions in a receding-horizon belief tree, where action values are measured by finite-horizon expected task-completion cost rather than local search utility alone. To keep online planning tractable, the tree search uses multi-stage pruning and a terminal cost estimate for unfinished task events. When new receptacles are discovered, an embedding-based relevance gate limits LLM queries to updates that are likely to affect the target beliefs. Fig. 2 summarizes the proposed closed-loop framework.

This paper makes two contributions. First, we formulate online semantic exploration and temporal task execution as a hierarchical belief-space planning problem, where actions are selected according to their expected contribution to the remaining task. Second, we develop a tractable receding-horizon belief-tree planner that coordinates exploration and temporal task-execution decisions under target-location uncertainty. Experiments in hospital-like transport scenarios and a ProcTHOR simulation show that the proposed framework reduces execution cost and LLM-query overhead compared with baselines based on deterministic semantic instantiation and single-target model-based search.

II. PROBLEM FORMULATION

A. Semantic Environment Representation

We consider a partially known semantic environment in which the room-level topology is available before execution, while the receptacle-level structure is only partially known. At decision step t , the robot maintains a semantic map $\mathcal{M}_t = (\mathcal{R}, \mathcal{C}_t, \mathcal{E}_t)$, where $\mathcal{R}$ is the set of rooms, $\mathcal{C}_t$ is the set of currently known searchable receptacles, and $\mathcal{E}_t$ encodes room connectivity and known room-receptacle containment. The prior map is $\mathcal{M}_0 = (\mathcal{R}, \mathcal{C}_0, \mathcal{E}_0)$, with $\mathcal{C}_0 \subseteq \mathcal{C}_t$. A receptacle denotes any searchable entity that may contain or support a target, such as a cabinet, drawer, shelf, or open support surface. During exploration, newly observed receptacles are added to $\mathcal{C}_t$ and the containment relation in $\mathcal{E}_t$ is updated. Target locations are initially unknown, and each target is assumed to occupy one searchable receptacle in the semantic environment.

This hierarchy defines the granularity at which target-location uncertainty is represented. We instantiate a semantic location as $\ell = (r, c)$, where $r \in \mathcal{R}$ is a room and $c \in \mathcal{C}_{t,r}$ is a searchable receptacle currently known in that room. This representation separates room-level uncertainty from receptacle-level uncertainty within each room. For example, rather than treating “medicine in the pharmacy” as a single hypothesis, the model separates the probability that the medicine is in the pharmacy from the conditional probability that it is on a shelf, in a fridge, or inside a cabinet.

B. Problem Statement

We consider a high-level user command ξ that specifies N targets to be transported to desired destinations under optional temporal ordering constraints. The command defines the task targets $\mathcal{O} = \{o_1, \dots, o_N\}$, their delivery destinations $\mathcal{D} = \{d_1, \dots, d_N\}$, and precedence relations among task events. Target locations are unknown; each target must be localized before it can be transported. At decision step t , the robot execution state x_t records its semantic location and task status, including carried, localized, and delivered targets.

The objective is to find an online policy π that minimizes the expected task makespan,

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T_\pi} c(s_t, a_t) \right],$$

where s_t is the full planning state, $a_t = \pi(s_t)$, $c(s_t, a_t)$ includes navigation and symbolic operation time, and T_π is the stopping time when all task requirements are satisfied. The challenge is that observations and task-execution decisions jointly change the remaining task: an observation can alter future search priorities, while an execution action can change the value of later observations. The robot must therefore choose actions by their expected effect on completing the remaining temporal task.

III. METHODS

Our framework selects actions in belief space by their expected effect on the remaining temporal task, following the closed-loop overview in Fig. 2.

A. Temporal Task Specification

We use a temporal task interface to convert a user command into a structure that can be used by online planning. A high-level command ξ is first represented as a temporal specification, following language-based temporal-task interfaces [6]. As in automaton- and LTL-based task decomposition methods [7], the specification is then compiled into a partial-order task graph. We denote this graph by a DAG $\mathcal{G}_\xi = (\mathcal{V}_\xi, \mathcal{E}_\xi)$, where vertices are required task events and directed edges encode precedence constraints. During execution, the planner maintains an execution state q on $\mathcal{G}_\xi$, which records satisfied event preconditions and determines which task events are currently admissible.

B. Hierarchical Semantic Belief

Objects in human environments exhibit regularities in where they are stored or used. Following recent LLM-guided belief-estimation work [3], we use an LLM to initialize common-sense priors over target locations; action selection is handled by the belief-space planner. For each target o_i , we maintain a belief over semantic locations $\ell = (r, c)$, where $r \in \mathcal{R}$ is a room and $c \in \mathcal{C}_{t,r}$ is a searchable receptacle currently known in that room. The belief is factorized as

$$b_i(\ell) = b_i(r, c) = P(o_i \in r)P(o_i \in c \mid o_i \in r).$$

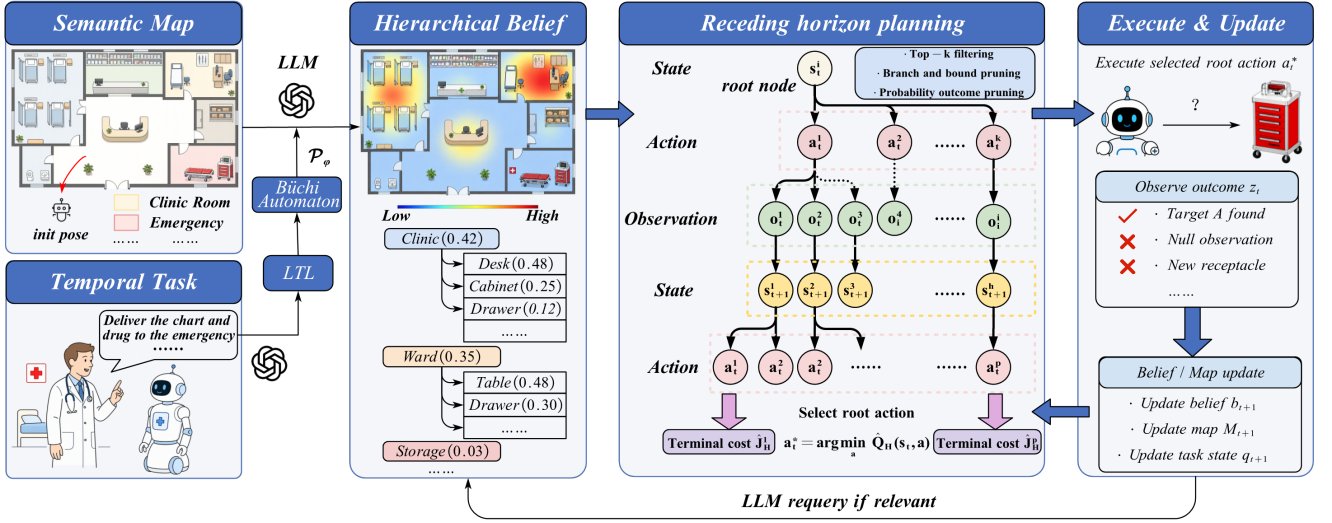


Fig. 2. Overview of the proposed framework. A user command and a partial semantic map are used to construct temporal task constraints and LLM-guided hierarchical beliefs over target locations. The robot evaluates exploration and execution actions with receding-horizon belief-tree search, executes the selected root action, and updates the belief, semantic map, and task state from observations.

This factorization quantifies target-location uncertainty at two semantic levels: a room-level distribution and a receptacle-level conditional distribution within each room.

After an exploration action, the belief is updated over the room–receptacle semantic locations. Let $\mathcal{S}_a$ denote the set of semantic locations inspected by exploration action a . For target o_i , the update is

$$b'_i(\ell) = \begin{cases} \mathbb{I}[\ell = \ell^{\text{obs}}], & \text{if } o_i \text{ is detected at } \ell^{\text{obs}}, \\ \eta b_i(\ell) p_{\text{fn}}, & \text{if } o_i \text{ is not detected and } \ell \in \mathcal{S}_a, \\ \eta b_i(\ell), & \text{if } o_i \text{ is not detected and } \ell \notin \mathcal{S}_a, \end{cases}$$

where p_{fn} is the false-negative probability and η normalizes the distribution after a negative observation. The hierarchical factors are then obtained by marginalization:

$$P(o_i \in r) = \sum_{c \in \mathcal{C}_{t,r}} b'_i(r, c), P(o_i \in c \mid o_i \in r) = \frac{b'_i(r, c)}{P(o_i \in r)},$$

for rooms with nonzero marginal probability.

When exploration discovers a new receptacle $\tilde{c}$ in room r , the belief support is expanded to include the new semantic location $(r, \tilde{c})$. Since querying the LLM after every discovery is costly and not every receptacle is informative for every target, we use an embedding-based relevance gate. Given a text embedding model $e(\cdot)$, we compute $\gamma_{i,\tilde{c}} = \cos(e(o_i), e(\tilde{c}))$ between each target o_i and the newly discovered receptacle $\tilde{c}$. If $\max_i \gamma_{i,\tilde{c}} > \tau$, the receptacle is treated as task-relevant, and the LLM is queried to update the conditional beliefs over the expanded set $\mathcal{C}_{t,r}$. Otherwise, $\tilde{c}$ receives a small default probability mass.

C. Belief-Space Action Evaluation

At each decision step, the planner reasons over the robot execution state, hierarchical semantic belief, and task-graph execution state. The planning state is $s = (x, b, q)$, where x

is the robot execution state, $b = \{b_i\}_{i=1}^N$ is the hierarchical semantic belief over target locations, and q is the execution state on the partial-order task graph. The feasible action set $\mathcal{A}(s)$ contains task-execution actions enabled by the partial-order constraints and exploration actions grounded to semantic locations in the current map and belief.

Inspired by model-based search over possible outcomes [2], we evaluate candidate actions by combining immediate execution cost with estimated downstream task cost. Unlike single-target search, the value here is computed over a state that also includes task progress and feasible task-execution events. Let $c(s, a)$ denote the immediate action cost, including navigation and operation time. For action a , let $\mathcal{Z}(s, a)$ denote the possible outcomes, $P(z \mid s, a)$ the outcome probability under the current belief and observation model, and $\mathcal{T}(s, a, z)$ the successor state. We write the belief-space action value as

$$Q(s, a) = c(s, a) + \hat{\mathbb{E}}[J(s') \mid s, a],$$

where the second term estimates the downstream cost over possible observations and successor states. Computing this term exactly would require expanding future action–observation contingencies, which is too expensive for online execution. We therefore approximate it using the receding-horizon belief-tree search described in the next subsection.

As a result, a location with lower semantic probability may still be preferred if searching it is expected to reduce future exploration or task-execution cost, letting the planner coordinate exploration and execution online rather than committing to the most likely semantic location.

D. Receding-Horizon Belief-Tree Search

We approximate the downstream cost term with a receding-horizon belief-tree search. At each decision step, the search expands a finite-horizon action–observation tree rooted at the current state. Action nodes correspond to feasible robot

decisions, and observation branches represent possible exploration outcomes, such as detecting a target or receiving a negative observation. Each branch updates the semantic belief, robot execution state, and task-graph execution state, thereby changing the remaining planning problem.

Fully expanding this tree would provide complete lookahead over candidate actions, but doing so is computationally prohibitive for online planning in partially known environments. The number of tree nodes grows rapidly with the planning horizon because exploration actions may induce multiple observations, each of which changes later task-event choices. We therefore control tree growth through top- K action filtering, outcome pruning, cost-bound pruning, and horizon truncation.

Let $\mathcal{A}_{\text{exp}}(s)$ and $\mathcal{A}_{\text{exec}}(s)$ denote exploration and task-execution actions, respectively. All enabled execution actions are retained, while only the top- K exploration actions are kept according to a one-step score $\psi(s, a)$ that balances target belief mass and immediate action cost:

$$\tilde{\mathcal{A}}(s) = \mathcal{A}_{\text{exec}}(s) \cup \text{TopK}_K(\mathcal{A}_{\text{exp}}(s), \psi(s, \cdot)),$$

where $\text{TopK}_K(\mathcal{A}, f)$ denotes the K actions in $\mathcal{A}$ with the highest scores $f(a)$. For each retained action, outcome branches with probability below ϵ_z are discarded, and branches whose accumulated cost exceeds the current incumbent are pruned. These rules focus computation on the contingencies most relevant to root-action selection.

The pruned belief tree is expanded to depth H . At the cutoff, unfinished task events are evaluated by a terminal cost estimate. We first ground each target to a representative semantic location:

$$\hat{\ell}_i(s) = \begin{cases} \ell_i^{\text{obs}}, & o_i \text{ has been localized,} \\ \arg \max_{\ell} b_i(\ell), & \text{otherwise.} \end{cases}$$

Let $\hat{\mathcal{L}}(s) = \{\hat{\ell}_i(s)\}_{i=1}^N$. Given these grounded locations, the residual task is evaluated on the remaining partial-order task graph using a cost-aware branch-and-bound topological search. The resulting cost-to-go, denoted by $\hat{J}_{\text{term}}(s)$, provides a lightweight estimate of the remaining makespan beyond depth H .

We denote the resulting approximate horizon- H action value by $\hat{Q}_H(s, a)$. At decision step t , the robot constructs $s_t = (x_t, b_t, q_t)$, evaluates the pruned horizon- H belief tree, and selects

$$a_t^* = \arg \min_{a \in \tilde{\mathcal{A}}(s_t)} \hat{Q}_H(s_t, a).$$

Only the selected root action is executed before replanning. After execution, the robot incorporates the resulting observation, updates the semantic map, hierarchical belief, and task-graph execution state, and replans from the updated state. This closed-loop procedure avoids committing to a long action sequence whose assumptions may become invalid after new observations, and continues until all task requirements are satisfied.

Algorithm 1 summarizes the proposed closed-loop planner. The robot initializes target-location beliefs from LLM-guided

Algorithm 1 Receding-Horizon Belief-Space Planner

Require: prior semantic map $\mathcal{M}_0$, task DAG $\mathcal{G}_{\xi}$, horizon H , filtering size K , thresholds ϵ_z, τ , default mass ϵ_{new}

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1:  $x \leftarrow x_0, q \leftarrow q_0, b \leftarrow \text{LLMPRIOR}(\mathcal{M}_0, \mathcal{G}_{\xi})$ 
2:  $s \leftarrow (x, b, q), \mathcal{M} \leftarrow \mathcal{M}_0$ 
3: while  $q$  is not terminal do
4:    $(\mathcal{A}_{\text{exec}}, \mathcal{A}_{\text{exp}}) \leftarrow \text{GROUND ACTIONS}(s, \mathcal{M}, \mathcal{G}_{\xi})$ 
5:    $\tilde{\mathcal{A}}(s) \leftarrow \mathcal{A}_{\text{exec}} \cup \text{TopK}_K(\mathcal{A}_{\text{exp}}, \psi(s, \cdot))$ 
6:   for all  $a \in \tilde{\mathcal{A}}(s)$  do
7:      $\hat{Q}_H(s, a) \leftarrow \text{BELIEFTREEBACKUP}(s, a, H, \epsilon_z, \hat{J}_{\text{term}})$ 
8:   end for
9:    $a^* \leftarrow \arg \min_{a \in \tilde{\mathcal{A}}(s)} \hat{Q}_H(s, a)$ 
10:  Execute  $a^*$  and observe  $z$ 
11:   $(\mathcal{M}, b, x, q) \leftarrow \text{UPDATE}(\mathcal{M}, b, x, q, a^*, z)$ 
12:  for all new receptacles  $\tilde{c}$  observed in  $z$  do
13:     $\gamma_{i, \tilde{c}} \leftarrow \cos(e(o_i), e(\tilde{c})), \quad i = 1, \dots, N$ 
14:    if  $\max_i \gamma_{i, \tilde{c}} > \tau$  then
15:       $b \leftarrow \text{LLMUPDATE}(b, \tilde{c}, \mathcal{M})$ 
16:    else
17:       $b \leftarrow \text{ASSIGNDEFAULTMASS}(b, \tilde{c}, \epsilon_{\text{new}})$ 
18:    end if
19:  end for
20:   $s \leftarrow (x, b, q)$ 
21: end while
```

semantic priors and repeatedly constructs a filtered action set $\tilde{\mathcal{A}}(s)$. Candidate root actions are evaluated by a pruned horizon- H belief-tree backup with terminal cost estimation, and only the minimizer of $\hat{Q}_H$ is executed. After each observation, the robot updates the semantic map, belief, execution state, and task-graph state; newly discovered receptacles trigger LLM updates only when they pass the embedding-based relevance gate. The process repeats until the task graph reaches a terminal state.

IV. EXPERIMENTS

This section evaluates the proposed framework in hospital-like transport scenarios and a ProcTHOR simulation. All algorithms were implemented in Python 3 and tested on a desktop computer with an Intel Core i7-14700KF CPU.

A. Experimental Setup

The hospital benchmark uses a partially known room-receptacle hierarchy; Fig. 3 shows a representative layout. Room connectivity is known, but only part of the receptacle hierarchy is initially available, with additional receptacles discovered during exploration. We evaluate temporal transport tasks with $N \in \{3, 5, 7\}$ targets over 17 randomized trials per scale. Targets and true receptacle locations are sampled from the semantic prior over candidate receptacles, and all methods use the same task instances, ground-truth placements, prior maps, and cost, observation, LLM, and embedding models.

We report execution time (Exec.), travel distance (Dist.), exploration actions (Exp.), negative observations (Neg.), online decision cost (Plan+LLM), and total target-location entropy

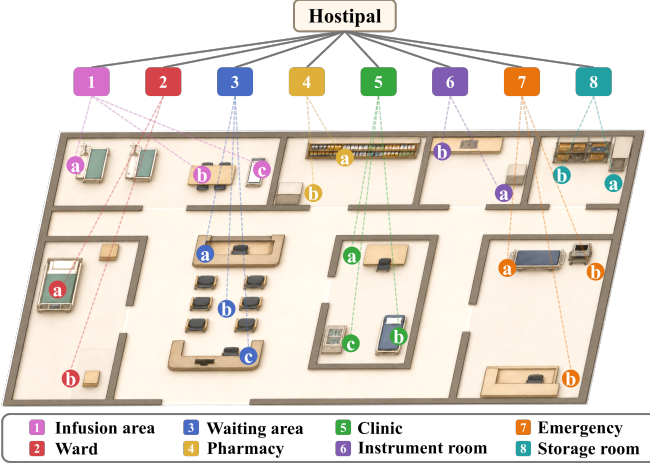


Fig. 3. An example of the hospital-like hierarchical semantic environment.

$H_{\text{tot}}(t) = -\sum_{o_i \in \mathcal{O}} \sum_{\ell} b_{i,t}(\ell) \log b_{i,t}(\ell)$. Exec. includes navigation and symbolic operation time, while Plan+LLM reports pure planning time plus total LLM-query latency, with average LLM calls in parentheses. We use Qwen2.5-7B-Instruct [8] as the local LLM and multilingual-e5-small [9] as the embedding model. Unless otherwise stated, our method uses $H=3$, $K=5$, $p_{\text{fin}}=0.1$, $\epsilon_z=0.05$, $\tau=0.5$, and $\epsilon_{\text{new}}=0.05$.

B. Baselines and Simulation

We compare our method with two planning baselines.

a) *EPoG-style global planner*: Following EPoG [5], each unlocalized target is instantiated at its most likely semantic location under the LLM prior, and the resulting deterministic task graph is solved by the same cost-aware topological search used in our terminal estimator. When execution contradicts the instantiated hypothesis, the semantic map and target hypotheses are updated and replanning is triggered, but newly discovered receptacles are only incorporated reactively, without a separate exploration-action interface.

b) *LLM-informed model-based search (LLM-MB)*: Adapted from the single-target planner of [2], this baseline evaluates search actions independently for each target and selects the locally best one, while deliveries are planned on the same task DAG and cost model, and new receptacles are handled by the same embedding-based relevance gate as our method. It thus preserves object-search-level lookahead, but its value estimate remains target-local and ignores the global route and later precedence-constrained events.

Fig. 4 illustrates the decision behavior of the three planners on one hospital transport instance, with executed trajectories over the belief heatmaps (a)–(c) and the corresponding entropy profiles (d)–(f). The planners differ first in how they resolve target-location uncertainty at the beginning of task execution. Our planner searches the nearby ward-side region, where the drug and the chart are not the most likely hypotheses but are semantically plausible and close to the start, reducing uncertainty at low cost before committing to a farther high-probability location such as the pharmacy. The EPoG-style

TABLE I
MAIN COMPARISON RESULTS ACROSS METHODS AND TASK SCALES.
VALUES ARE AVERAGED OVER 17 RANDOMIZED TRIALS.

N	Method	Exec. (s)	Dist. (m)	Exp.	Neg.	Plan+LLM (s)
3	EPoG-style	56.73	116.74	/	1.94	0.07 +3.03 (1.94)
	LLM-MB	60.26	122.17	5.35	1.47	0.13+2.71 (1.11)
	Ours	45.36	93.97	4.24	0.94	0.23+ 1.59 (0.88)
5	EPoG-style	97.42	190.36	/	3.30	0.18 +4.97 (3.30)
	LLM-MB	104.18	212.61	8.47	3.65	0.47+3.98 (2.24)
	Ours	85.85	171.43	7.12	2.18	0.72+2.68 (1.65)
	Ours($H=4$)	78.62	158.91	6.71	1.94	1.93+ 2.49 (1.59)
7	EPoG-style	126.60	263.58	/	5.84	0.36 +7.04 (5.84)
	LLM-MB	139.52	292.74	13.06	6.35	1.07+6.21 (5.06)
	Ours	96.40	207.86	9.15	2.72	1.69+ 3.93 (3.18)

planner instead visits the strongest semantic hypotheses first, so when the true location differs from its single instantiation it incurs early detours and corrective replanning. LLM-MB also reasons about search uncertainty, but its target-local value can favor a search that helps one object without improving the multi-target route.

As exploration proceeds, the decision focus shifts from reducing uncertainty to weighing further search against task progress. Once the nearby targets are resolved, our belief-tree planner implicitly compares the expected task-completion cost of delivering immediately against that of continuing to search for the remaining targets, and switches to execution when additional observations are unlikely to lower the remaining task cost, a route- and ordering-aware trade-off that the target-local baselines do not make.

We further instantiate the framework in a lightweight high-fidelity ProcTHOR simulation [10], whose semantic rooms, receptacles, and object placements form the planner’s map. Given a command of the same form as in the hospital scenario, the robot runs the full receding-horizon belief-space loop without modification, and Fig. 5 shows the resulting execution trace in first-person and top-down views.

C. Main Comparison

Table I reports the main comparison across the three task scales. With its default horizon $H=3$, our method attains the lowest execution time and travel distance at every scale (at $N=5$ the deeper $H=4$ variant is best overall), reducing execution time by 18.6% and 24.4% over the EPoG-style planner and LLM-MB and travel distance by 16.9% and 23.8% on average over the three scales. The margin also widens with scale: the execution-time gap over the EPoG-style planner grows from 11.4s at $N=3$ to 30.2s at $N=7$.

This advantage originates in the exploration statistics. Scoring each action by its expected effect on the remaining task, rather than by local search likelihood, yields far fewer negative observations: at $N=7$ our method incurs 2.72 failed searches on average, against 5.84 and 6.35 for the two baselines, which directly shortens the executed route and explains the lower travel distance. The EPoG-style planner has no separate exploration-action interface, so its exploration count is omitted

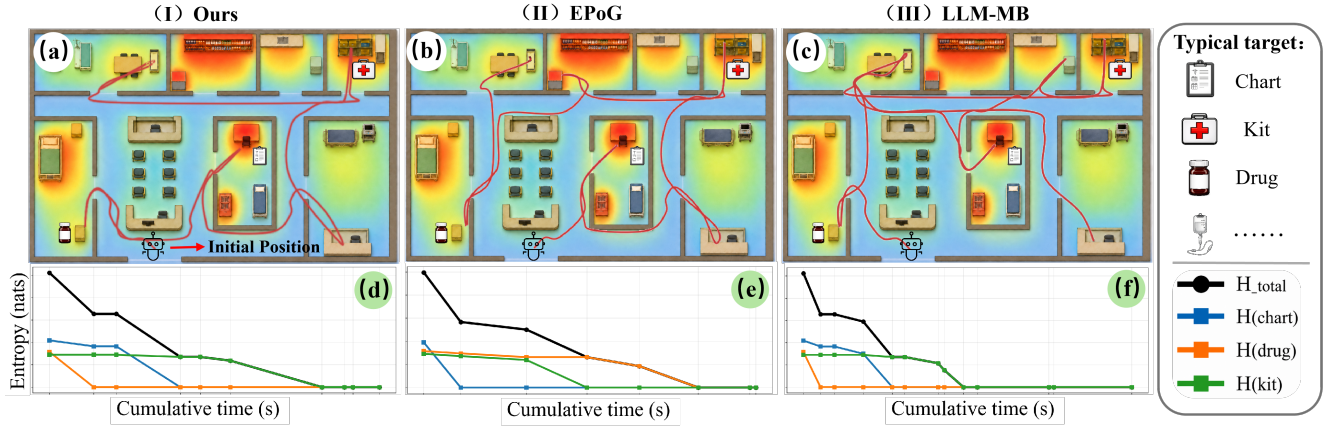


Fig. 4. Decision behavior of the three planners on one hospital transport instance. (a)–(c) Executed trajectories of (I) our method, (II) the EPoG-style planner, and (III) LLM-MB, overlaid on heatmaps of the semantic target-location belief (warmer colors indicate higher probability). (d)–(f) The corresponding total and per-target belief-entropy profiles over cumulative execution time.

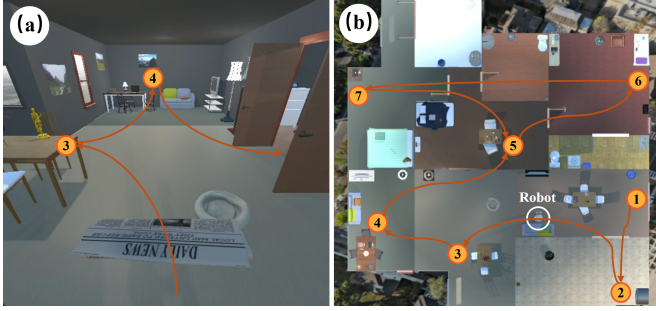


Fig. 5. Execution of the proposed planner in a ProcTHOR scene. (a) First-person view with the overlaid navigation trajectory. (b) Top-down floor plan used for semantic mapping and planning, with the executed waypoint sequence.

(marked ‘/’); its high negative-observation count reflects the cost of committing to a single most-likely instantiation that observations frequently invalidate. Consistent with this, the entropy profiles in Fig. 4(d)–(f) show that our method drives the total target-location entropy down to near zero with the least execution time, reflecting more efficient uncertainty resolution.

The online decision cost reveals the trade-off. Our method spends the most pure planning time, raising it at $N=7$ from 0.36s to 1.69s, but this suppresses redundant LLM queries: average LLM calls drop from 5.84 and 5.06 to 3.18, cutting LLM latency to 3.93s from 7.04s and 6.21s. The combined Plan+LLM time is therefore the smallest among all methods (5.62s at $N=7$). The $H=4$ variant at $N=5$ confirms the same trade-off: it shortens execution time and travel distance by 8.4% and 7.3%, but raises pure planning time about $2.7\times$ (from 0.72s to 1.93s), so we adopt $H=3$ as a practical default.

V. CONCLUSION

We presented an LLM-guided hierarchical belief-planning framework that coordinates exploration and temporal task execution in partially known environments by maintaining

target-location beliefs, updating them from observations, and evaluating actions by expected downstream task cost through receding-horizon belief-tree search. Experiments in hospital-like scenarios and ProcTHOR show reduced execution cost and LLM-query overhead compared with deterministic semantic instantiation and single-target model-based search baselines. Future work will consider richer perceptual uncertainty, larger environments, and multi-robot settings.

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